

⇒ Computer

A computer is a machine that can solve problems by accepting data, performing certain operations and presenting the results of those operations under the detail direction of step-by-step instructions.

⇒ Program

Such set of sequenced instructions which cause a computer to perform particular operations are called program.

⇒ Computer Science

Computer Science is the study of operating principles of a computer, computer programming languages and algorithms for solving theoretical as well as practical problems.

It involves

- Development and use of devices for processing Information.

⇒ Data

Information in one form (raw data) is presented as Input to the computer and in another form (processed data / results) are required from the computer at its output.

⇒ Data Processing

Data processing is transforming raw material of information into useful information by computer.

⇒ Characteristics of a computer

- 1) Speed
- 2) Accuracy
- 3) Reliability
- 4) Storage Capability
- 5) Reduced Cost
- 6) Intangible Benefits

Classification of computers according to size

⇒ Large Computer Systems

- 1) Super Computer
- 2) Mainframe
- 3) Mini Computers

Small Computer Systems

- 1) Micro Computer
- 2) Laptop / Notebooks
- 3) Hand-Held Computer
- 4) Pen-Driven Computers

⇒ Super Computers.

When we talk about speed, Super Computers are the biggest and fastest computers - in terms of speed of processing data.

SuperComputers are designed such that they can execute trillion of instructions per seconds.

This is because of multiple interconnected processors in super computers which perform instructions / process in parallel - This is called parallel processing.

Scientific and Engineering applications:

- 1) Weather Forecasting
- 2) Scientific Simulation
- 3) Nuclear Energy Research

Developed by Roger Cray in 1976.

⇒ Main frame Computer

⇒ expensive Computer

⇒ high storage capacity

⇒ Great performance

⇒ MIPS

⇒ runs smoothly and has long life

3) Mini computer

⇒ low weight

⇒ less expensive

⇒ fast

⇒ 4-200 users

⇒ parallel processing

⇒ two or more processors

4) Workstation

⇒ single user

⇒ technical - scientific applications

⇒ expensive

⇒ focuses on one specific task

⇒ complex work purpose

⇒ large storage capacity

⇒ animation, Data Analysis, CAD

audio-video creation and editing

5) PC

1) limited number of softwares can be used

2) smallest in size

3) easy to use

4) personal use

5) microprocessor as CPU.

6) Analogue Computer

continuous data that changes continuously & cannot have discrete value is analogue data

7) Digital Computer

8) Hybrid

⇒ Looking Inside a Computer

There are four parts of computer system

1) Hardware

3) Users

2) Software

4) Data

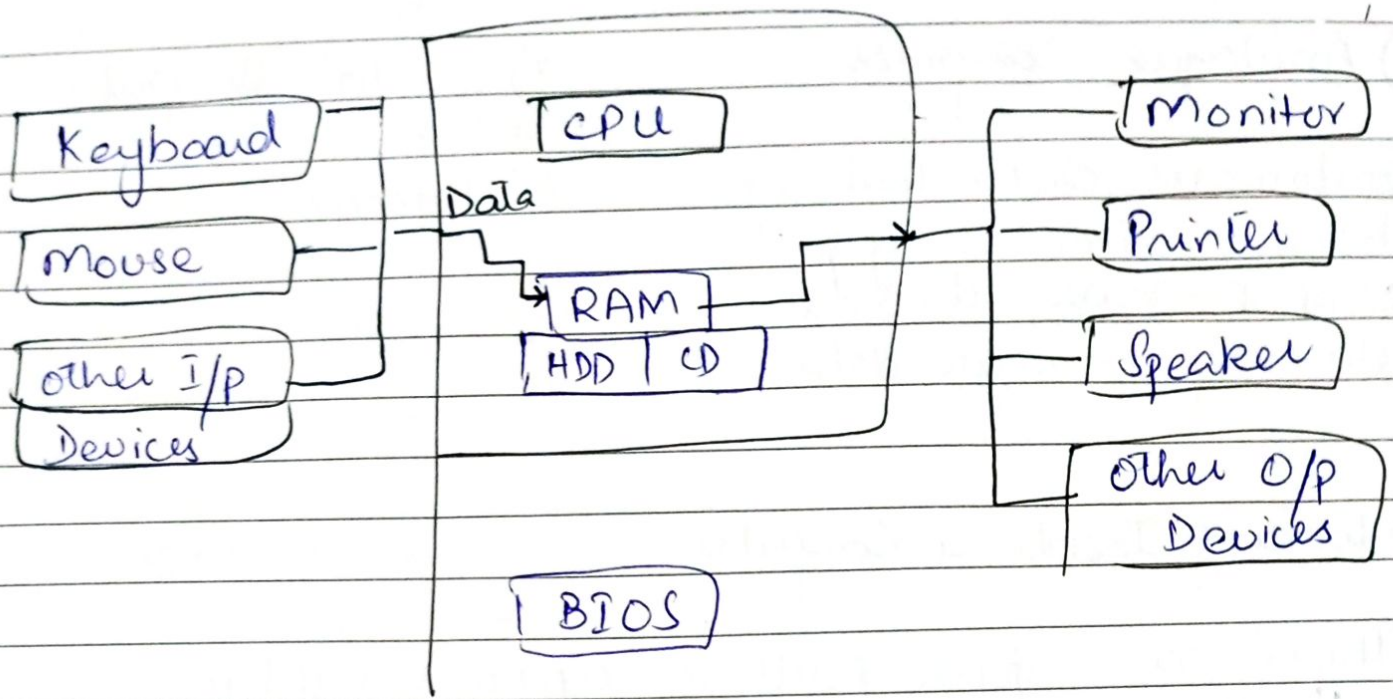
Hardware are mechanical parts, any thing that can be touched, consists of combination of electronic devices that can be used to control computer operation, Input, output

⇒ Device ⇒ hardware

Software ⇒ Program

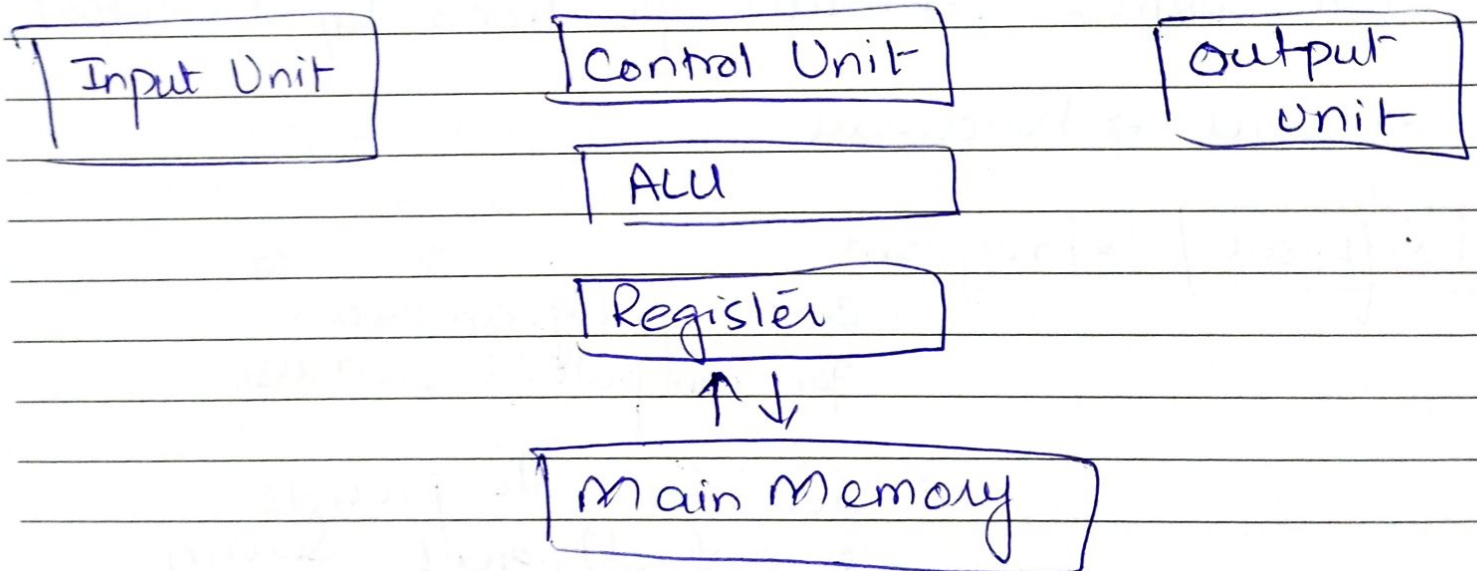
⇒ Some programs are for computer's own use

⇒ Some are to provide several different services to users.



Von Neumann Architecture

John Von Neumann $\Rightarrow$ 1945



⇒ Central Processing Unit is the electronic circuit responsible for executing the instructions of a computer program

⇒ Sometimes referred as MicroProcessor or processor.

⇒ It consists of the following parts

⇒ Register

⇒ Registers are high speed storage devices

⇒ All the data must be stored in some register before it can be processed

MAR	⇒ holds the address of the data to be accessed
MDR	⇒ hold data that is to be transferred to or from memory
AC	⇒ perform arithmetic or logic
PC	⇒ Next instruction
CIR	⇒ current instruction